FROM ANSWERS TO POSSIBILITIES

How Young Professionals Develop Problem-Framing and Judgement Skills Through Human–AI Collaboration

EMANAIRE DC13 | Technical University of Munich | Research proposal

1 Background and research gap

Across spring and winter university intakes, I have mentored, trained and worked with interns and recent graduates entering professional life. I have become curious about how some approach unfamiliar challenges: they begin with a desired outcome, use digital and increasingly AI-enabled tools to explore routes towards it, introduce constraints and revise their questions along the way. This is a practitioner observation, not a claim about an entire generation. It prompts me to ask whether generative AI is changing not only the answers young professionals can produce but also how they learn to frame problems and exercise judgement.

Research on knowledge workers suggests that GenAI can shift critical-thinking practices towards verification and integrating outputs (Lee et al., 2025). Work on human–AI co-creation indicates that iterative exchanges do not automatically improve joint creativity; how ideas are developed together matters (Luan et al., 2025). Yet the developmental question remains important: when young professionals produce more sophisticated outputs with AI, which problem-framing and judgement capabilities—if any—are they developing themselves, and can those capabilities transfer to unfamiliar work?

2 Aim and research questions

The aim is to explore how iterative human–AI collaboration relates to the development of problem-framing and professional judgement among early-career knowledge workers. The leading question is: How does collaboration with generative AI shape how young professionals frame unfamiliar workplace problems, explore possibilities and exercise independent judgement?

RQ1 How do participants use AI dialogue to define and reframe problems, generate alternatives and incorporate real-world constraints?

RQ2 How do they distribute cognitive work with AI, assess its suggestions, detect limitations and decide when human judgement should take precedence?

RQ3 How do these practices change with experience, and to what extent is their reasoning transferable to related tasks with less AI assistance?

3 Conceptual framing

I would connect problem framing and reframing (identifying underlying needs and questioning a task's initial definition) with human–AI co-agency (negotiating contributions and retaining meaningful responsibility) and professional capability development. I would distinguish observable process quality—questions asked, alternatives compared, assumptions tested—from final output quality. I do not assume that AI necessarily improves or weakens thinking: it may support exploration in some situations while substituting for it in others.

4 Proposed empirical approach

A staged, exploratory mixed-method design could begin with interviews and walkthroughs of anonymised AI-assisted tasks, recording how early-career professionals formulate, revise and evaluate their questions. In a participatory element, young professionals could help identify authentic workplace scenarios rather than merely complete researcher-designed tasks, drawing on Cortesi, Hasse and Gasser's (2021) work on meaningful youth participation (while adapting its principles to an adult workplace population).

Findings would inform comparable scenario-based tasks with and without AI support, examining the relevance of problem definitions, breadth of feasible alternatives, treatment of constraints, justification of choices, detection of errors and appropriate delegation. Follow-up tasks could probe whether reasoning changes over time and transfers to new situations. Sampling across roles and workplace contexts would be considered, alongside factors such as task difficulty, prior AI experience, access to mentoring, time pressure and organisational expectations. Consent, confidentiality, privacy of workplace materials and limits of causal inference would guide the design. The precise comparisons, measures and feasibility would be refined with the supervisors.

5 Expected contribution

The study could offer (i) a more precise account of problem framing as a human–AI collaboration skill; (ii) behavioural evidence distinguishing effective AI-assisted results from skills demonstrated by the person; and (iii) implications for AI fluency, early-career learning and workplace practices that support judgement, appropriate delegation and autonomy. Its contribution would be to examine a developmental mechanism rather than presume that AI-native familiarity itself constitutes professional competence.

6 Research fit and collaborative development

My 12+ years in enterprise technology, product management and organisational transformation, together with coaching interns and graduates and experimenting with AI-enabled workflows, give me access to concrete questions about capability and work. My small exploratory study, When AI Thinks With Us, reinforced my interest in moving from observation to rigorous empirical inquiry. As a first-generation university graduate who learned much through self-directed exploration, I also care deeply about access to guidance and opportunities to develop independent judgement.

Prof. Sandra Cortesi's youth-centred, participatory scholarship particularly resonates with my intention to investigate young professionals' actual practices rather than impose deficit assumptions. I would welcome the opportunity to develop the constructs, methods and research scope with Prof. Cortesi and the supervisory team, and to let participants' experiences inform the research as it progresses.

Selected references

Cortesi, S., Hasse, A., & Gasser, U. (2021). Youth participation in a digital world: Designing and implementing spaces, programs, and methodologies. Berkman Klein Center. https://doi.org/10.2139/ssrn.3844115

Lee, H.-P., et al. (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. CHI 2025. https://doi.org/10.1145/3706598.3713778

Luan, Y. L., Kim, Y. J., & Zhou, J. (2025). Augmented learning for joint creativity in human–GenAI co-creation. Information Systems Research. https://doi.org/10.1287/isre.2024.0984